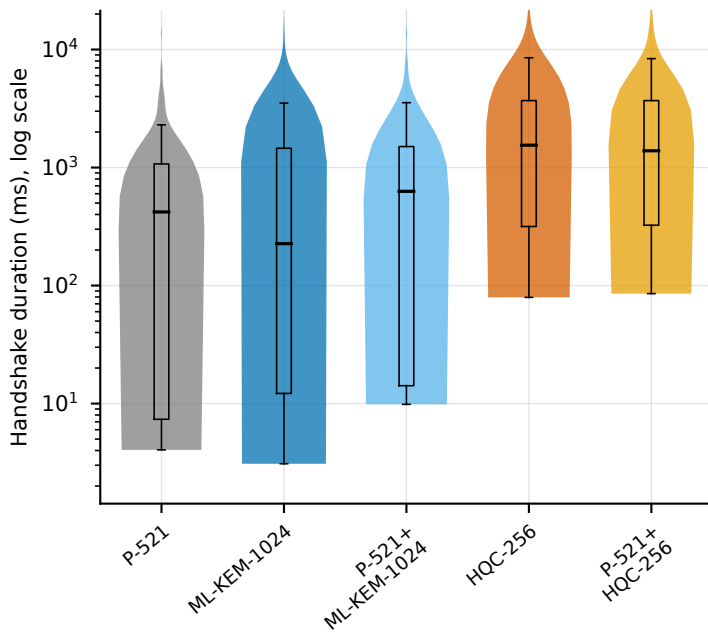
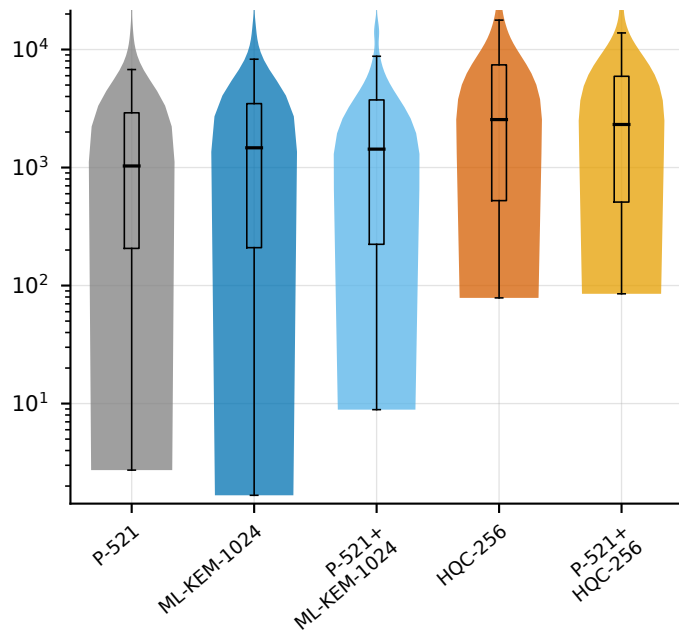


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	secp521r1 Fail	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)	mldsa87 Fail
P-521	1.11k	219.9	27.82k	1/500	3.56k	275.1	110.11k	1/500
ML-KEM-1024	2.30k	416.5	110.29k	1/500	3.96k	252.0	134.25k	3/500
P-521+ML-KEM-1024	1.73k	255.0	52.31k	1/500	3.46k	179.3	64.09k	3/500
HQC-256	5.54k	273.9	114.42k	4/500	7.71k	198.6	122.69k	9/500
P-521+HQC-256	4.61k	256.0	148.31k	0	7.15k	223.4	119.40k	5/500